Name: ____________________________________

Use the Triangle Sum Theorem.

1. Two angles of a triangle measure 43° and 78°. Find the third angle.

2. The angles of a triangle measure (2x)°, (3x + 5)°, and (4x − 14)°. Find x and all three angles.

Use the Exterior Angle Theorem.

3. An exterior angle measures 142°. One remote interior angle measures 65°. Find the other.

4. An exterior angle measures (4x + 6)° and its remote interior angles measure 58° and 64°. Find x.

5. An exterior angle of a triangle measures 135° and one remote interior angle measures 62°. Find the other remote interior angle AND the interior angle adjacent to the exterior angle.

Use the corollaries.

6. The acute angles of a right triangle measure (3x + 7)° and (2x + 8)°. Find x and both angles.

7. Each angle of an equiangular triangle is written as (6y)°. Find y.

Answer each question.

8. Classify a triangle whose sides measure 7, 7, and 10 and whose angles measure 40°, 40°, and 100°.

9. In two triangles, ∠A ≅ ∠D and ∠B ≅ ∠E. If m∠A = 52° and m∠B = 61°, find m∠F. Name the theorem you used.

10. Can a triangle have two right angles? Explain using the Triangle Sum Theorem.